

Airline Reservation System

1. Introduction

Airline Reservation System

The Airline Reservation System is a SQL-based database project developed to manage flight information, customer details, seat availability, and flight bookings. The system provides a structured way to store and retrieve airline reservation data using a relational database.

The project focuses on designing a normalized database schema and implementing SQL operations such as data insertion, flight searching, seat availability checking, booking management, and cancellation. Triggers and views are also used to automate booking-related updates and generate useful reports.

2. Abstract

The **Airline Reservation System** is designed to efficiently manage airline flight and reservation information using **MySQL**. The database consists of four main tables: **Flights**, **Customers**, **Seats**, and **Bookings**.

The system maintains relationships between these tables using primary keys and foreign keys and applies constraints to maintain data integrity. Sample flight, customer, seat, and booking records were inserted into the database.

SQL queries were implemented to search for flights and identify available seats. Triggers were created to automatically update seat status when bookings are confirmed or cancelled. Two views were also created to display flight availability and booking summary information.

The project demonstrates how a relational database can be used to organize and manage an airline reservation system effectively.

3. Tools Used

- **MySQL Workbench** – Database development and SQL execution
- **MySQL** – Relational Database Management System
- **SQL** – Database creation, manipulation, queries, triggers, and views

4. Steps Involved in Building the Project

Step 1: Database and Schema Creation

A database named `airline_reservation` was created. Four main tables were designed:

- **Flights** – stores flight number, airline, source, destination, departure/arrival time, and fare.
- **Customers** – stores customer name, email, and phone number.

- **Seats** – stores flight seats, seat number, class, and availability status.
- **Bookings** – stores customer bookings, selected seats, booking status, and total amount.

Step 2: Constraints and Relationships

Primary keys, foreign keys, NOT NULL, and UNIQUE constraints were implemented to maintain data integrity and establish relationships between the tables.

Step 3: Sample Data

Sample records were inserted into the Flights, Customers, Seats, and Bookings tables for testing the system.

Step 4: SQL Queries

SQL queries were implemented for:

- Searching flights based on source and destination.
- Displaying available seats.
- Displaying booking information.
- Updating booking status.
- Retrieving booking details using joins.

Step 5: Triggers

Triggers were created to automatically manage seat status when a booking is inserted, confirmed, or cancelled. This helps keep booking and seat availability information synchronized.

Step 6: Views and Reports

Two views were created:

- **vw_flight_availability** – displays total seats, available seats, and booked seats for flights.
- **vw_booking_summary** – provides a combined booking report containing customer, flight, seat, booking, and payment information.

5. Conclusion

The Airline Reservation System successfully demonstrates the use of a relational database for managing airline reservations. The project implements database design, normalization, constraints, sample data, SQL queries, triggers, and views.

The system provides an organized method for managing flights, customers, seats, and bookings while automatically maintaining seat availability. Through this project, practical knowledge of **SQL and MySQL database management** was gained, including database design, relationships, joins, triggers, views, and data manipulation.